The Copperbelt University

Kapasa Makasa University Campus

Department of Agriculture and Aquatic Sciences

SESSIONAL EXAMINATIONS 2023

COURSE CODE: NR 430

COURSE NAME: PLANT PROTECTION

TIME ALLOWED: THREE (3) HOURS

DATE: 23rd OCTOBER 2023

TIME: 09:00hrs

INSTRUCTIONS:
1. This paper comprises of seven (7) questions
2. Answer the COMPULSORY QUESTION ONE
3. Answer any TWO questions from Section B
4. Answer any TWO questions from Section C
5. You are required to answer a total of five (5) questions in the exam
6. Each question carries 20 marks

CamScanner

SECTION A

QUESTION 1 (compulsory)

(a) Explain various parameters used in classification and identification of plant diseases **(12 marks)**

(b) Explain how Fluorescence In-Situ Hybridization (FISH) works as a molecular method of disease identification. Further state the advantages of this molecular method. **(8 marks)**

SECTION B

QUESTION 2

(a) In details, discuss plant disease tetrahedron **(8 marks)**

(b) Define the following terms

i. Stridulation
ii. Phytopathogenesis
iii. Biotrophs
iv. Phytoanticipins **(8 marks)**

(c) Describe various distinct stages involved in pathogenesis **(4 marks)**

QUESTION 3

(a) Insects are living organisms that belong to the class *insecta* and phylum *athropoda*. State and describe five (5) ''*Orders*' into which insects can be classified including examples of insects that belong to such *Orders*. **(10 marks)**

(b) Briefly explain how bacteria reproduce **(5 marks)**

(c) Describe the differences between viruses and viroids **(5 marks)**

CamScanner

JESTION 4

(a) Pests are known to develop resistance to a certain pesticide after a given period of time. Discuss how this phenomenon occur **(5 marks)**

(b) Explain how plants defend themselves against infections using pre-existing defense structures **(10 marks)**

(c) Nematodes are known to cause a number of problems and diseases in plants. Discuss the damages caused by nematodes and how they cause such damages **(5 marks)**

SECTION C

QUESTION 5

(a) Discuss similarities and differences between compound interest diseases and simple interest diseases. **(10 marks)**

(b) Briefly, explain how Polymerase Chain Reaction (PCR) works as a plant disease identification method. **(5 marks)**

(c) List various symptoms and signs caused by plant bacterial diseases. **(5 marks)**

QUESTION 6

(a) Discuss the four-tiered approach that anchors Integrated Pest Management (IPM). **(8 marks)**

(b) Explain in details how viruses invade the plant and cause diseases **(4 marks)**

(C) Briefly explain various characteristics of arthropods **(8 marks)**

QUESTION 7

(a) Explain various cultural practices that aim at reducing the adverse effects of plant diseases. **(10 marks)**

(b) In details, explain the importance of understanding insect classifications as an Agroforester **(10 marks)**

C I P

-Spread rapidly
-pores are spread by
air

Simple

- Spread slowly
- pores are spread by mud, animal
- have a long incubation period

CamScanner

THE COPPERBELT UNIVERSITY

PLANT PROTECTION TEST 2

Instructions:

In this question paper, there is one (1) questions portioned in four (4) task areas and you are required to attempt all the questions in two (2) hours

Q1. Use an academic method to critique the following International Conventions:

a) Framework Convention on Climate Change **[25 Marks]**

b) Convention on Biological Diversity **[25 Marks]**

c) Convention to Combat Desertification **[25 Marks]**

d) Convention on International Trade in Endangered Species of Wild Fauna and Flora **[25 Marks]**

END OF TEST

KAPASA MAKASA UNIVERSITY

PLANT PROTECTION –NR 430

TEST 1

DATE: 22ND MARCH, 2022

DURATION: 1 HR, 30 MIN.

INSTRUCTIONS:

Answer **ALL** the questions in this paper

Use separate answer sheets

1. What do you understand by the following terms
   a. Virulence is the disease producing power of an organism
   b. Epiphytes plants which grow on other in order to obtain physical support
   c. Facultative saprophytes organisms that live in/on dead organic matter (fungi & bacteria)
   d. Pathogenesis process by an infectious agent when it comes in contact with an host.
   e. Phyto-anticipins these are pre-existing toxins found in plant cells and tissues that defend the plant from phytopathogens. **(10 marks)**

2. In details, discuss ''Disease Tetrahedron''. **(12 marks)**

3. State various methods that can be used to diagnose mollicutes infection in plants. **(4 marks)**

4. Describe how plants defend themselves against diseases using Induced Structural defence mechanisms. **(10 marks)**
   - layer of abscion cells

5. When is the plant considered to be healthy? **(4 marks)**
   - able to carry out photosynthesis
   - able to absorb nutrients
   - able to travel mineral salts
   - able to reproduce

Total= 40 marks

THE END..

ELISA

- Pathogenesis is the process by which a disease or disorder develops.

The Copperbelt University

Kapasa Makasa University Campus

Department of Agriculture and Aquatic Sciences

SESSIONAL EXAMINATIONS 2022

COURSE CODE: NR 430

COURSE NAME: PLANT PROTECTION

TIME ALLOWED: THREE (3) HOURS

DATE: 19 TH OCTOBER 2022

INSTRUCTIONS:

1. Answer the COMPULSORY QUESTION ONE
2. Answer ONE other question from Section A to make it TWO
3. Answer at least ONE question from Section B or C
3. You are required to answer a total of five (5) questions in the exam as follows.

   i) 2 from section A including the COMPULSORY question one

   ii) At least ONE from BOTH section B and C

   iii) You are at liberty to pick ONE OR TWO from section A and Section B

CamScanner

NR 430 PLANT PROTECTION

SECTION A (Answer Question 1 and ONE other question from this section)

Question One (COMPULSORY) ✓

a) Explain various parameters used in classification and identification of plant diseases.
Affected plant type
Type of symptoms Affected organ

(12 marks)

b) Describe the various and distinct stages involved in pathogenesis. (Pre-penetration, penetration, post penetration, inoculation) **(6 marks)**

c) What is dieback? this is when leaves start dying from the top. **(2 marks)**

Question Two ✓

a) Discuss disease tetrahedron in detail. **(12 marks)**

b) Define the following terms:

i. Virulence producing power of an organism

ii. Phytopathogenesis its the process by which plants become infected by pathogens and how respond to the infection

iii. Biotrophs organisms the feed on living tissues

iv. Phytoanticipins these are pre-existing toxins found in plant cells that help to fight the plant from phytopathogens. **(8 marks)**

Question Three

a) Explain the various cultural practices that aim at reducing the adverse effects of plant diseases.
rotation, quarantine & isolation, weeding, inter cropping, Following, disease resistant seeds, disease free materials, disinfecting tools, biosecurity **(10 marks)**

b) Briefly explain how bacteria reproduce. **(5 marks)**

c) Explain the difference between viruses and viroids. **(5 marks)**

- both are agents that can cause diseases in plants but they differ in different ways.
- viruses are larger in size while viroids are smaller and simpler
- viruses can replicate only within living cells while viroids can replicate within plant cells
- viruses can travel through many means such as nematodes while viroids can travel through infected plant tissues.
- viruses are viroids differ in size, structure transmission, replication strategy, impact on plant health.

SECTION B (Answer one or two questions from this section)

Question Four

a) Pests are known to develop resistance to a certain pesticide after a certain period of time. Discuss how this phenomenon occurs. ***(5 marks)***

b) Expalin how plants defend themselves against infections using pre-existing defense structures. presence of hairy structures, presence of thorns, thick cuticle and epidermis quantity of wax, shape, size of stomata. ***(10 marks)***

c) Explain how Enzyme Linked Immuno-sorbent Assay **(ELISA)** works as a molecular method of disease identification. ***(5 marks)***

Question Five

a) Describe four (4) major plant viral diseases and how they affect a given plant species. ***(10 marks)***

b) Briefly explain how Polymerase Chain Reaction **(PCR)** works as a plant disease identification method. ***(5 marks)***

c) List the various symptoms and signs caused by plant bacterial diseases. ***(5 marks)***

- wilting
- cankers
- stunted growth
- necrosis
- root rot
- soft rot

Section C (Answer one or two questions from this section)

Question Six

a) Discuss the various principles that anchor Integrated Pest Management (IPM). ***(8 marks)***
Efficiency, economic cost, practicability, accessibility

b) Briefly explain various characteristics of arthropods. ***(8 marks)***
Segmented body parts, paired appendages, bilateral symmetry, exoskeleton

c) List various cultural practices that can be incorporated into an IPM system. ***(4 marks)***
Pest resistant crops, weeding, proper application of fertilizers, inter cropping

Question Seven

a) Insects are living organisms that belong to the Class *Insecta* and Phylum *Anthropoda*. State and explain five (5) insect "Orders" into which insects can be classified including examples of insect order you have selected. ***(10 marks)***

b) Give a detailed explanation of the importance of understanding insect classification as an Agro-forester. ***(10 marks)***

CamScanner